

Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Summary	Recently, our organization experienced a DoS attack which compromised the internal network for 2 hours. Our network services suddenly stopped responding due to a flood of ICMP packets, normal internal network traffic could not access any network resources due to this. The incident management team responded by blocking incoming ICMP packets, stopping all non-critical network services offline, and restoring critical network services. The cybersecurity team then investigated the event and found that a malicious actor had sent a flood of ICMP pings into the company's network through an unconfigured firewall. This vulnerability allowed the malicious actor to overwhelm the internal network through a denial of service [DoS] attack. Steps were then taken by the cybersecurity team to improve the company's network security.
Identify	Upon investigation the cybersecurity team found that a malicious actor had found a vulnerability and was successful in launching a denial of service [DoS] attack. The company's network was flooded with ICMP pings through an unconfigured firewall which caused the internal network to be overwhelmed and unable to allow normal network traffic to access network resources for 2 hours.

Protect	Due to findings during the investigation of this recent security incident, the cybersecurity team has established a plan to improve the company's network security. A new firewall is in place with a configuration to limit the rate of incoming ICMP packets. An additional new configuration on the firewall is source IP address verification to check for spoofed IP addresses on incoming ICMP packets. A security information and event management {SIEM} tool application will also be utilized to detect abnormal traffic patterns. An IDS/IPS system will also be used to filter out some ICMP traffic based on suspicious characteristics.
Detect	The company's cybersecurity team has implemented the following changes in order to prevent such security attacks in the future: The firewall has newly added configurations to include a limit on the rate of incoming ICMP packets and a source IP address verification. A security information and event management {SIEM} tool will be utilized to detect abnormal traffic patterns. Also an IDS/IPS system will be used to filter out some ICMP traffic based on suspicious characteristics.
Respond	The incident management team responded by blocking incoming ICMP packets, stopping all non-critical network services offline, and restoring network services. All customers that notified us they were unable to use network services during the 2 hours the network was compromised due to the attack were emailed coupons and discount codes for the inconvenience.
Recover	The team has performed a thorough investigation of the security incident, and followed up with an internal audit of the network security strategy and made some improvements based on what was learned from the event and discovered during the internal audit. The cybersecurity team put in place changes that will make future attacks from malicious actors much more difficult and will continue to audit and make adjustments on a regular basis to prevent future security incidents.

Reflections/Notes: